

Sovereign European AI as the Heart of the ANThERA Ecosystem

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Executive Summary

ANTHERA

APERTUS

ANTHERA Systems selects Apertus — the open-source Large Language Model developed by **ETH Zurich + EPFL + CSCS** (Swiss National Supercomputing Centre) — as the **default AI engine** for all products in its ecosystem, starting with **GOVERN AI**.

This is not a technical choice; it is a **strategic and identity-defining** one: European data stays in Europe, the intelligence processing it is European, and the entire AI supply chain is transparent, inspectable and compliant.

In one sentence: *"European data deserves European AI."*

APERTUS STRATEGY

1. Why Apertus — The ANThERA Rationale

1.1 The current problem

European enterprises adopting AI face a dilemma:

Requirement	US LLM (GPT/Claude)	Consequence
GDPR personal data	Extra-EU transfer	Complex Data Processing Addendum, Schrems III risk
DORA data sovereignty	Critical outsourcing	Authority notification, contingency plan
EU AI Act Art. 13/53 transparency	Secret training data	Impossible to attest provenance
NIS2 resilience	US supplier dependency	Geopolitical single point of failure
Public sector procurement	US Cloud Act	Exclusion from strategic tenders

1.2 The ANThERA answer

"European data deserves European AI"

Apertus resolves all of these frictions **in one shot**:

- **100% transparent** (public weights + training data + recipe)
- **Sovereign EU/CH** (trained on Alps supercomputer in Lugano, EU hosting)
- **Natively multilingual** (Italian, German, French, Romansh on par with English)
- **Apache 2.0** (zero vendor lock-in, free for commercial use)
- **EU AI Act aligned by design** (not retrofit)

1.3 The ANThERA commercial message

"Your data is European. Your governance is European. The AI processing it must be European too."

This narrative becomes a **sales multiplier**:

- Italian banks (Bank of Italy + ECB requirements)
- Public Administration (AGID procurement, sovereignty)
- Healthcare (patient data, GDPR Art. 9)
- Defence / critical infrastructure (NIS2 supplier chain)

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- Public sector CH/EU (Public AI Inference Utility by Swisscom)

2. What Apertus Is — Technical Datasheet

2.1 Identity

Attribute	Value
Name	Apertus (Latin for "open")
Developers	ETH Zurich, EPFL, CSCS (Swiss National Supercomputing Centre)
Sponsor	Swiss Confederation, Swiss AI Initiative
Release	September 2025
License	Apache 2.0 (commercial-friendly)
Site	apertus.ai
Available models	Apertus-8B-Instruct, Apertus-70B-Instruct
Supported languages	1,800+ (Italian, German, French, Romansh natively)
Trained on	Alps supercomputer (CSCS Lugano) — 10,000+ NVIDIA GH200
Transparency	Weights + dataset + training recipe + code fully public
Compliance	EU AI Act Article 13/53 ready, GDPR-friendly

2.2 Distinctive features

- **Fully Open**: it is the first enterprise-scale LLM where **everything** is public (not only the weights)
- **Balanced multilingual**: unlike GPT/Claude (90% English), Apertus treats Italian and German as first-class languages
- **Auditability**: every model decision is traceable to training data
- **Geopolitical independence**: outside the perimeter of US Cloud Act, CHIPS Act, extraterritorial sanctions

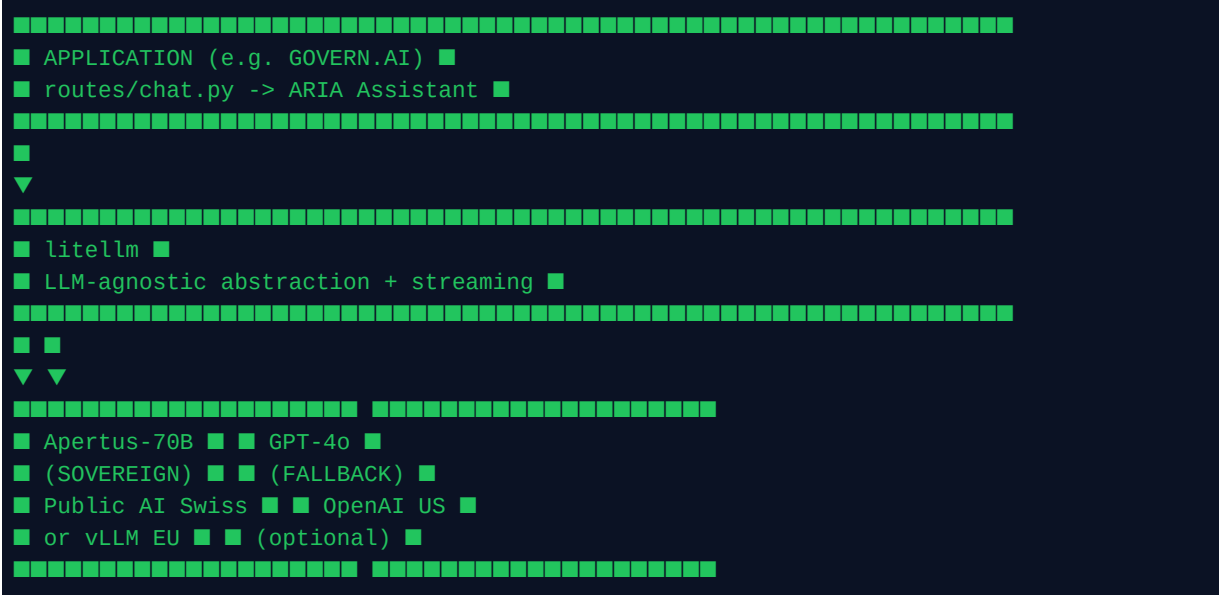
2.3 Positioning vs competition

Feature	GPT-4o	Claude 3.5	Gemini 2.0	Apertus 70B
License	Proprietary	Proprietary	Proprietary	Apache 2.0
Public weights	No	No	No	Yes
Public training data	No	No	No	Yes
EU/CH sovereignty	No (US)	No (US)	No (US)	Yes (CH)
First-class IT multilingual	Partial	Partial	Partial	Yes
EU AI Act Art. 13 ready	Complex	Complex	Complex	Native
US Cloud Act	Yes	Yes	Yes	No
Self-hosting	No	No	No	Yes
Cost	High	High	Medium	Free (infra cost only)

3. ANTHERA Integration Architecture

3.1 Principle: abstraction layer

ANTHERA already uses `litellm` across all products as an **LLM-agnostic abstraction layer**. This makes Apertus adoption **transparent** to the application code.



3.2 Switch via environment variable

Integration is limited to:

```
# backend/.env
LLM_PROVIDER=apertus
LLM_MODEL=apertus/apertus-70b-instruct
LLM_BASE_URL=https://api.publicai.swiss/v1
APERTUS_API_KEY=<key>
```

Zero application code changes. Routing happens at the `litellm` layer.

3.3 Suggested operating mode (Dual Mode)

To maximize commercial flexibility, ANTHERA suggests exposing a **"Sovereign Mode" toggle** in the UI:

Mode	Model	Use case
Sovereign Mode (default)	Apertus 70B	Production, sensitive data, EU/CH clients
Performance Mode (opt-in)	GPT-4o / Claude	Very complex tasks, prototypes
Offline Mode	Apertus 8B via Ollama	On-prem demo, air-gapped environments

This becomes a **killer sales argument**: *"Want total sovereignty? Sovereign Mode. Want max quality? Performance Mode. Your choice, in 1 click."*

4. Deployment Options — Step by Step

4.1 Option A — Public AI Inference Utility (Swisscom) RECOMMENDED

Scenario: SaaS production, zero ops, declared CH sovereignty.

Steps:

- Register on publicai.swiss (Swiss Public AI Inference Utility, beta)
- Obtain API key

- Configure:

```
LLM_MODEL=apertus/apertus-70b-instruct
LLM_BASE_URL=https://api.publicai.swiss/v1
APERTUS_API_KEY=<key>
```

- Smoke-test endpoint with curl
- Restart backend

Time: ~2 hours (incl. Swisscom onboarding)

Cost: per-token pricing, in line with OpenAI but CH-sovereign

Pros: zero infra, enterprise SLA, certified CH/EU compliance

Cons: single-provider dependency (mitigated by CH public entity)

4.2 Option B — Hugging Face Inference Endpoints

Scenario: rapid prototyping, customer pilot.

Steps:

- Account on huggingface.co
- Deploy "Apertus-70B-Instruct" on Inference Endpoints (EU region: Frankfurt or Helsinki)
- Obtain endpoint URL + token
- Configure:

```
LLM_MODEL=huggingface/apertus-70b-instruct
LLM_BASE_URL=https://<endpoint>.eu-west-1.aws.endpoints.huggingface.cloud
HUGGINGFACE_API_KEY=<token>
```

Time: ~3 hours

Cost: ~2-4 USD/hour of active A100 GPU (auto-scaling)

Pros: deploy in 10 minutes, EU infra

Cons: HF is US-based corp (mitigated: EU hosting + Apache license on the model)

4.3 Option C — Self-Hosted vLLM on EU GPU MAX SOVEREIGNTY

Scenario: enterprise, banks, public administration, sensitive clients.

Steps:

- EU GPU provisioning: OVH (Strasbourg/Roubaix), Aruba (Italy), Open Telekom (DE), Scaleway (FR), or on-prem
- Minimum hardware: 2x NVIDIA H100 80GB (for 70B), 1x A100 40GB (for 8B)
- vLLM installation:

```
pip install vllm
python -m vllm.entrypoints.openai.api_server \
--model swiss-ai/Apertus-70B-Instruct-2509 \
--port 8000 \
--tensor-parallel-size 2
```

- Configure GOVERN.AI:

```
LLM_MODEL=openai/apertus-70b-instruct
LLM_BASE_URL=http://<vllm-host>:8000/v1
OPENAI_API_KEY=dummy # vLLM accepts any key
```

Time: 1-2 days (incl. GPU setup + benchmark)

Cost: ~3,000-8,000 EUR/month dedicated H100 (or CAPEX ~50k EUR on-prem)

Pros: total sovereignty, zero external dependency, optimal latency

Cons: ops overhead, GPU capex/opex

4.4 Option D — Ollama On-Prem (Apertus 8B)

Scenario: local demo, developers, air-gapped environments.

Steps:

- Install Ollama (`brew install ollama` or Linux equivalent)
- `ollama pull apertus:8b`
- `ollama serve` (port 11434)
- Configure:

```
LLM_MODEL=ollama/apertus:8b
LLM_BASE_URL=http://localhost:11434/v1
```

Time: 30 minutes

Cost: zero (runs on a MacBook M3 with 32 GB RAM)

Pros: offline demo, zero cost, maximum privacy

Cons: 8B only (lower quality), not scalable for production

5. ANTHERA Adoption Roadmap

Phase 1 — GOVERN.AI (Q2 2026)

- Integrate Apertus as a configurable option in GOVERN.AI (~1 sprint)
- "Sovereign Mode" toggle in Settings UI
- Update documentation + Investor Intro: "Powered by Sovereign European AI"
- Pilot demo with an IT banking customer with Sovereign Mode active

Phase 2 — Production validation (Q3 2026)

- Select 1 enterprise design partner for Apertus 70B pilot on Public AI Swiss
- Measure: latency, ARIA response quality, user satisfaction
- Blind A/B comparison vs GPT-4o
- Case study publication

Phase 3 — Extension to the entire ANTHERA ecosystem (Q4 2026 — Q1 2027)

- Apertus as default on every new ANTHERA product
- Retrofit existing products where applicable
- Formal partnership with Swiss AI Initiative / CSCS
- ANTHERA as "Apertus Premier Partner" (proposed title)

Phase 4 — Sovereignty go-to-market (2027)

- CH market opening (Swisscom, UBS, legacy Credit Suisse, federal PA)
 - DE/AT market opening (BaFin, Bundesbank, Vienna Stock Exchange)
 - FR market opening (CNIL, ACPR, defence sector)
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6. Compliance — Apertus vs Regulations Mapping

Regulation	Article	How Apertus helps
EU AI Act	Art. 13 (Transparency)	Public training data -> immediate attestation
EU AI Act	Art. 53 (GPAI Models)	GPAI with complete technical documentation already available
GDPR	Art. 44 (Transfers)	EU/CH hosting -> no extra-EU transfer
GDPR	Art. 35 (DPIA)	Simplified DPIA thanks to model transparency
DORA	Art. 28 (ICT Third-party)	Critical EU/CH supplier approvable by the Board
NIS2	Art. 21 (Supply Chain)	Sovereign supply chain, not US-dependent
D.Lgs. 262	Internal Controls	Auditable model for Dirigente Preposto
ISO 42001	AI Management	Model transparency = easier lifecycle control
US Cloud Act	N/A	Apertus outside US jurisdiction

7. Total Cost of Ownership (TCO) — Example

Scenario: GOVERN.AI in production with 50 concurrent users, 100,000 LLM queries/month.

Item	GPT-4o (OpenAI)	Apertus 70B (Public AI CH)	Apertus 70B (Self-host vLLM EU)
Model cost	~3,000 EUR/month	~2,000 EUR/month	0 EUR/month
Infra cost	0	0	~5,000 EUR/month (2x H100)
Compliance cost	High (DPA, assessments)	Low	Minimal
Ops overhead	Low	Low	Medium
Monthly TCO	~3,000 EUR + compliance	~2,000 EUR	~5,000 EUR
Sovereignty TCO	Low	High	Maximum
Vendor lock-in	High	Medium	Zero

8. FAQ

8.1 Is Apertus really at GPT-4o level?

Apertus-70B is competitive with GPT-4-turbo / Claude 3.5 Sonnet on European multilingual benchmarks (Llama-3-Eval, EU-Truthful, ItaEval). It is slightly below GPT-4o on very complex reasoning tasks, but for ARIA use cases (regulatory Q&A, summary, guidance) the difference is marginal and offset by native multilingual quality.

8.2 How much does the technical migration cost?

Zero or near-zero, thanks to litellm. 1-2 hours of a developer are enough to test the Apertus endpoint and validate ARIA with representative prompts.

8.3 What if Public AI Swiss has downtime?

Litellm supports automatic fallback: configure Apertus as primary and GPT-4o as fallback. Zero impact on the end user.

8.4 Can I use Apertus to generate PDF/CSV?

Yes, exactly as today with GPT. Generation is handled on the ReportLab/Python side, not by the LLM. The LLM is only used for ARIA's regulatory reasoning.

8.5 Does Apertus respond correctly in Italian?

Yes, excellently. Unlike GPT (which internally translates from English), Apertus treats Italian as a first-class language in training. Italian stylistic and legal quality is superior.

8.6 Can we fine-tune on Italian regulations?

Yes, Apertus is Apache 2.0 and weights are public. We can LoRA fine-tune on Italian regulatory corpus (civil code, criminal code, GDPR, financial laws) to create *"ARIA Legal Italian Edition"*. Cost: ~5-10k EUR GPU + 1 week of work.

8.7 What if OpenAI launches a better model?

Litellm allows model choice **per request**. We lose nothing: we will always have the option to use the best of the market as opt-in, but the sovereign EU default remains Apertus.

8.8 Can Apertus be used in ANTHERA commercial products?

Yes, Apache 2.0 allows unlimited commercial use. No obligation to release ANTHERA code using Apertus.

9. Operational Next Steps

- **Public AI Swiss evaluation:** registration, Swisscom contact, test API key
 - **Smoke test on GOVERN.AI** (dev env): env vars switch, ARIA streaming validation
 - **Blind benchmark:** 20 regulatory prompts -> GPT-4o vs Apertus 70B -> quality scoring
 - **GO/NO-GO decision** by end of Q2 2026
 - **Production roll-out** Q3 2026 with "Sovereign Mode" UI toggle
 - **External communication:** ANTHERA website update, press release "ANTHERA adopts Apertus"
 - **ANTHERA portfolio extension** Q4 2026
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10. Conclusion

The adoption of **Apertus** is not a technical choice: it is an **identity-defining statement**.

ANTHERA Systems exists to bring digital sovereignty to Europe. An AI governance product running on US LLMs betrays its own mission. With Apertus, ANTHERA becomes **end-to-end European**:

- **European data** -> hosted in EU/CH
- **European code** -> compliance-first open source
- **European AI** -> Apertus, developed and hosted in Switzerland
- **European governance** -> EU AI Act / GDPR / DORA / NIS2 compliant by design

"Sovereign Control Plane for Enterprise AI, powered by Sovereign European AI."

This is the promise ANTHERA can keep — **starting today**.

Contacts

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Apertus references:

- apertus.ai
- huggingface.co/swiss-ai
- publicai.swiss (Swiss Public AI Inference Utility)
- ethz.ch / epfl.ch / cscs.ch

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